



NOVABANK WEB API

CLOUD MODERNIZATION
PROOF OF CONCEPT

CONTEXT AND REQUIREMENTS

- **Novabank wants to evaluate public cloud.**
They have asked CloudNation to develop a proof of concept to migrate and modernize a Web API
- Requirements
 - Financial Services: regulated, EU data residency requirement applies
 - Explicit availability target: 99.90%
 - Disaster Recovery targets: RPO 1 hr, RTO 4 hrs
 - Auditability: store and retain logs centrally for a defined term, control and restrict access

CURRENT STATE ASSESSMENT

Web API hosted in on-premise data center

- Single VM
- PostgreSQL database – assumed on a single VM
- Single environment – no separate dev/test
- No stated, evaluated availability target
- Logs kept locally

TARGET ARCHITECTURE (1)

Cloud platform – Azure

- Driven by skills available in the project team
- Deployment to Azure meets stated requirements
- Other cloud platforms may be evaluated later

Prefer Platform Services (PaaS) over Infrastructure (IaaS)

- Azure App Service and PostgreSQL Flexible Server
- Reduce operational overhead, improve security posture, built-in high availability features
- PostgreSQL server is VNET injected, it has no public endpoint
- Web App connects to database privately through VNET Integration

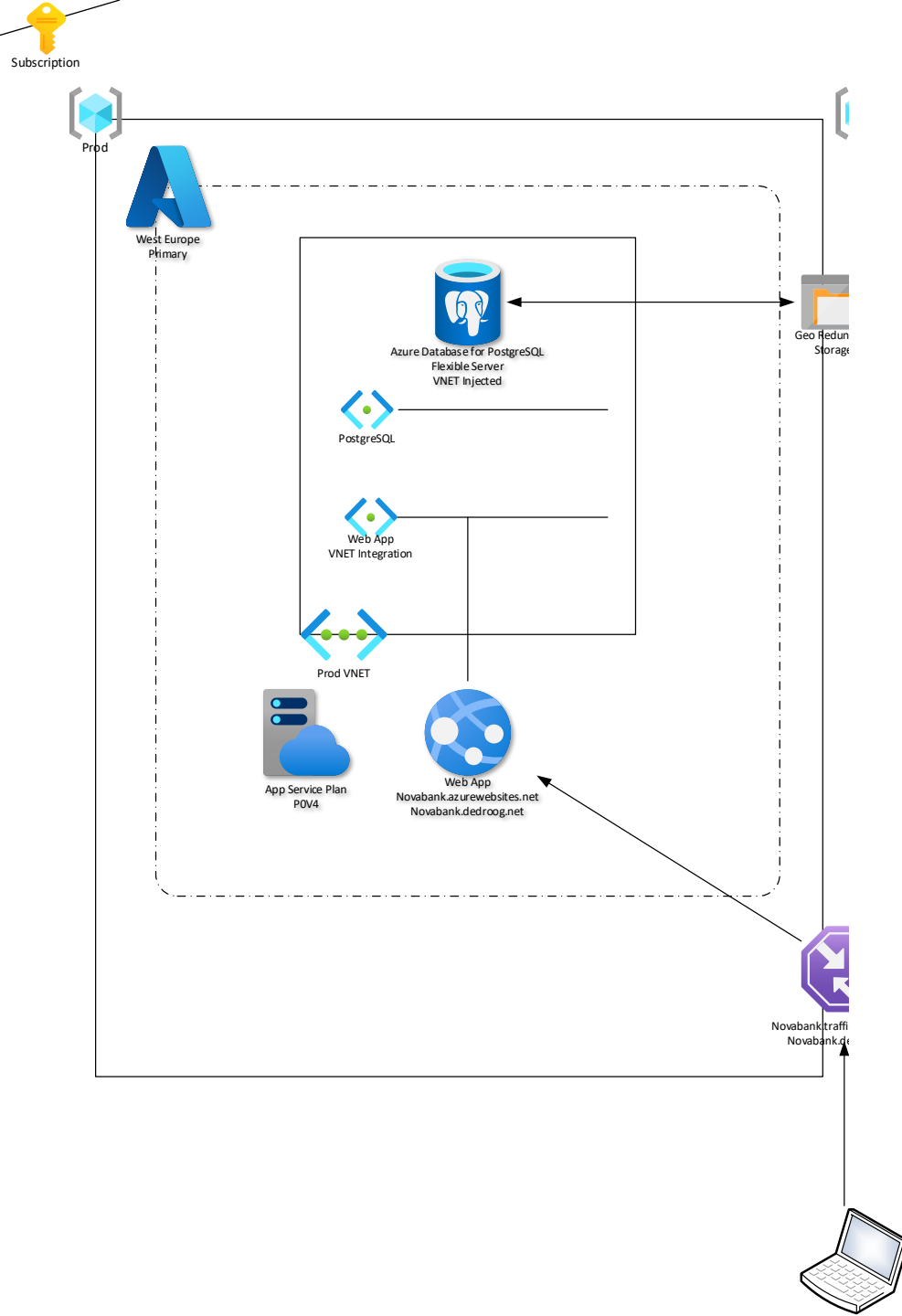
TARGET ARCHITECTURE (2)

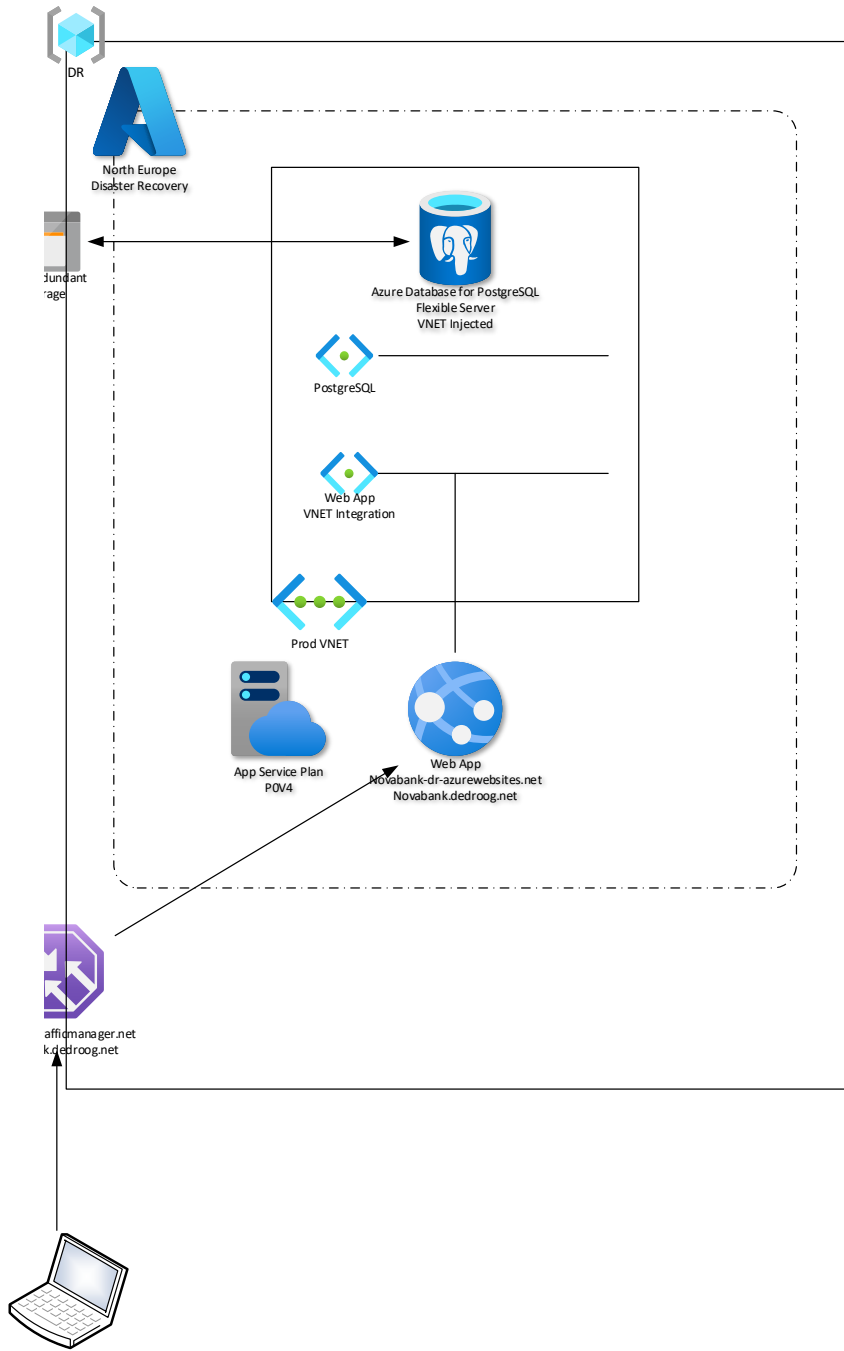
Primary – Disaster Recovery

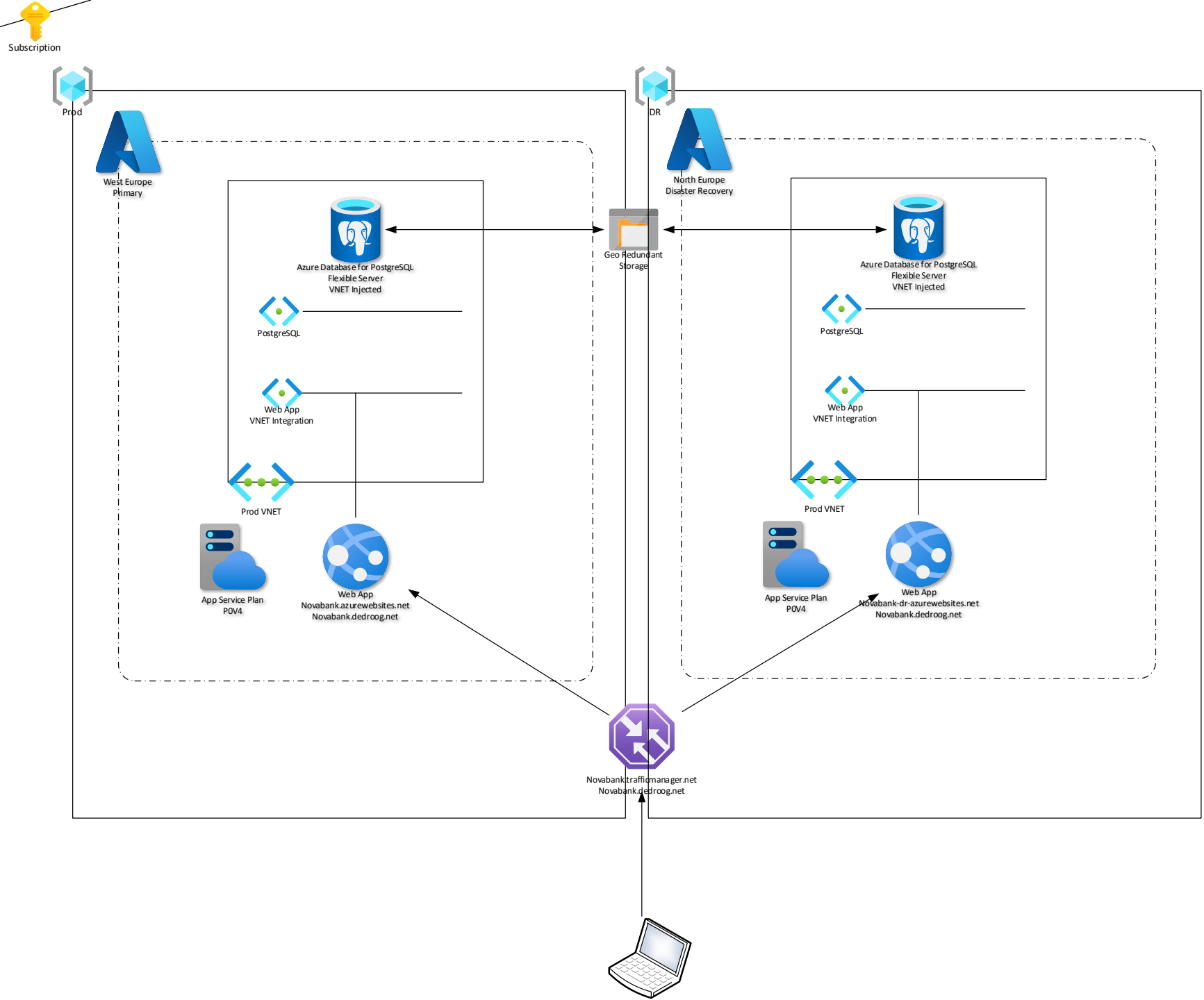
- Primary: West Europe, DR: North Europe
- DR is only deployed when Primary fails – no running costs when not needed
- PostgreSQL database configured for Geo Redundant backup
 - Data is copied asynchronously to North Europe, restored in DR
 - Database is restored in North Europe from copy when needed

Logging

- Logs sent to LogAnalytics Workspaces for each environment
- Retention set to 2 years
- Access to log content restricted to members of specific Entra Security Group with







DEVELOPMENT

Infrastructure as Code

- Bicep – leveraging Azure Verified Modules where possible
- Single code base, parameterized for primary, dr and dev deployments
 - deploymentMode = primary | dr
 - environmentMode = prod | dev
- No secrets in code or parameter files, password passed in from environment variable

No DNS reconfiguration in disaster situation

- Single, constant FQDN: Traffic Manager automatically routes traffic to live Web App

Primary – Disaster Recovery

- In disaster, just redeploy main.bicep with deploymentMode set to dr, no other actions needed
- Database is automatically restored from copy

Development environment

- Separate deployment
- Uses lower grade, cheaper component SKUs

PRACTICAL AI

AI-assisted engineering

- All Bicep code was developed in **Visual Studio Code with GitHub Copilot**
- Copilot accelerated code generation, refactoring, troubleshooting, and documentation
- Generated code remained subject to human review, Bicep validation, deployment testing, and architectural judgement

AI-assisted operations – options

- **IaC review agent:** Review pull requests for security, resilience, logging, tagging, and configuration gaps
- **Operations agent:** Correlate Azure Monitor and Log Analytics data and summarize incidents
- **Cost agent:** Identify underused capacity and recommend rightsizing, especially for development
- **DR assistant:** Guide operators through the recovery runbook and verify completion of each step
- **Knowledge agent:** Answer operational questions using approved architecture documentation and runbooks

AI supports engineers and operators, but production changes remain governed, auditable, and human-approved.

DEMO DEPLOYMENT

Web API application

- API component from YADA demo app [microsoft/YADA: Yet Another Demo App](#)
- Container image deployed to Web App by IaC, environment variables set appropriately
- To avoid certificate complexities, the demo environment uses non-TLS secured (http://) connections (this will be changed for production)
- API exposes endpoints to:
 - Query database for version
 - Return client source IP
 - Write client access events with source IP and timestamp to a table

Database

- Empty / default PostgreSQL deployment, no data loaded

TEST

Database version

http://novabank.dedroog.net/api/sqlversion

```
{
  "sql_output": "PostgreSQL 16.15 on x86_64-pc-linux-gnu, compiled by gcc (GCC) 13.2.0, 64-bit"
}
```

Client IP

http://novabank.dedroog.net/api/sqlsrcip

```
{
  "sql_output": "10.0.2.254"
}
```

Create table

http://novabank.dedroog.net//api/sqlsrcipinit

```
{
  "table_created": "srciplog"
}
```

Log access

http://novabank.dedroog.net//api/sqlsrciplog

```
{
  "srciplog": {
    "ip": "10.0.2.254",
    "timestamp": "2026-10-07 16:12:45.271077"
  }
}
```



THANK YOU

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